

The Transport Layer: TCP and UDP

The transport layer sits above the network layer in the TCP/IP model and provides end-to-end communication services for applications. Its two dominant protocols, the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP), offer very different trade-offs between reliability and speed. Both use 16-bit port numbers, ranging from 0 to 65535, to direct data to the correct application process on a host.

Transmission Control Protocol (TCP)

TCP is connection-oriented, reliable, and delivers a byte stream in order. Before any data is exchanged, the two endpoints establish a connection using a three-way handshake: the client sends a SYN segment, the server replies with a SYN-ACK, and the client confirms with an ACK. TCP guarantees reliable delivery by assigning a sequence number to each byte and requiring the receiver to return acknowledgments. If an acknowledgment is not received within a timeout, the sender retransmits the lost data. The standard TCP header is 20 bytes long.

Flow Control and Congestion Control

TCP uses a sliding window mechanism for flow control, which prevents a fast sender from overwhelming a slow receiver by limiting how much unacknowledged data may be in flight. Separately, congestion control prevents the sender from overwhelming the network itself. It begins with slow start, in which the congestion window grows exponentially until a threshold is reached, after which it enters congestion avoidance and grows linearly. The widely used strategy of increasing the window slowly and halving it on loss is known as additive-increase, multiplicative-decrease (AIMD).

User Datagram Protocol (UDP)

UDP is connectionless and unreliable: it sends independent packets called datagrams with no handshake, no acknowledgments, no retransmission, and no guarantee of ordering. This makes it faster and lower-overhead than TCP. The UDP header is only 8 bytes, compared with TCP's 20 bytes. Because it avoids the latency of connection setup and reliability machinery, UDP is preferred for applications that value speed over perfect delivery, such as the Domain Name System (DNS), live video and audio streaming, voice over IP (VoIP), and online gaming.

Choosing Between Them

The choice depends on application requirements. Use TCP when every byte must arrive correctly and in order—for example, web pages (HTTP), email, and file transfer. Use UDP when occasional loss is acceptable but low latency is critical. Some modern protocols, such as QUIC, build reliability features on top of UDP to get the best of both worlds.